Basic OSPF Configuration

Introduction

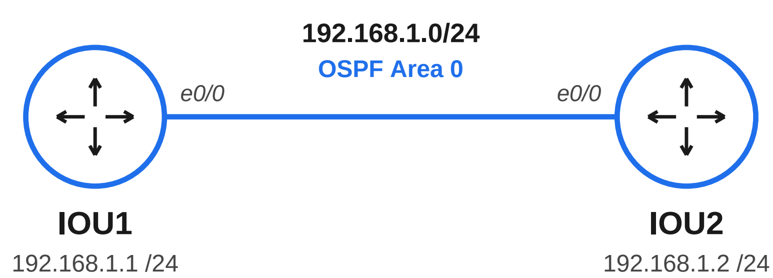
This lab demonstrates a minimal, two-router OSPF deployment. The topology consists of two Cisco IOU routers, IOU1 and IOU2, connected by a single Ethernet link. Both routers will be configured to run OSPF within a single area (Area 0) so that they form a neighbor adjacency and exchange routing information dynamically.

Objective

Configure IOU1 and IOU2 so that both routers establish an OSPF neighbor adjacency over the 192.168.1.0/24 network and can successfully ping one another's directly connected interface.

Lab Topology

The topology for this lab contains a single point-to-point Ethernet network, 192.168.1.0/24, connecting the e0/0 interface of IOU1 to the e0/0 interface of IOU2. OSPF Area 0 is enabled on this link so that both routers can dynamically learn and advertise routes to one another.



Command Summary

Command	Description
`configure terminal`	enters global configuration mode from privileged EXEC mode
`enable`	enters privileged EXEC mode
`end`	ends and exits configuration mode
`exit`	exits one level in the menu structure
`hostname host-name`	sets the device name
`interface type number`	changes from global configuration mode to interface configuration mode

Command	Description
`ip address ip-address subnet-mask`	assigns an IP address to an interface
`router ospf process-id`	enters router configuration mode for OSPF
`network network-address wildcard-mask area area-id`	enables OSPF on the matching interface(s) within the specified area
`ping ip-address`	sends an ICMP Echo Request to the specified address
`show ip ospf neighbor`	displays OSPF neighbor adjacency information
`show ip route`	displays the IP routing table
`show running-config`	displays the active configuration file
`shutdown; no shutdown`	disables an interface; enables an interface

IP Addresses

Device	Interface	IP Address	Subnet Mask
IOU1	Ethernet0/0	192.168.1.1	255.255.255.0
IOU2	Ethernet0/0	192.168.1.2	255.255.255.0

Lab Tasks

Task 1: Configure Router Interfaces

On each router, issue the commands required to configure a host name that matches the name of the router in the topology. Then configure the e0/0 interface with the IP address and subnet mask assigned to it in the IP Addresses table, and enable the interface.

Task 2: Configure OSPF

On each router, issue the commands required to start an OSPF process and advertise the connected 192.168.1.0/24 network into Area 0.

Task 3: Verify OSPF Neighbor Adjacency and Connectivity

On each router, issue the command to display OSPF neighbor information and confirm that an adjacency has formed with the other router. Then, from each router, ping the remote router's e0/0 interface to confirm connectivity.

Lab Solutions

Task 1: Configure Router Interfaces

On IOU1, issue the following commands:

```
Router>enable
Router#configure terminal
Router(config)#hostname IOU1
IOU1(config)#interface ethernet 0/0
IOU1(config-if)#ip address 192.168.1.1 255.255.255.0
IOU1(config-if)#no shutdown
```

On IOU2, issue the following commands:

```
Router>enable
Router#configure terminal
Router(config)#hostname IOU2
IOU2(config)#interface ethernet 0/0
IOU2(config-if)#ip address 192.168.1.2 255.255.255.0
IOU2(config-if)#no shutdown
```

Task 2: Configure OSPF

On IOU1, issue the following commands:

```
IOU1(config)#router ospf 1
IOU1(config-router)#network 192.168.1.0 0.0.0.255 area 0
```

On IOU2, issue the following commands:

```
IOU2(config)#router ospf 1
IOU2(config-router)#network 192.168.1.0 0.0.0.255 area 0
```

After both routers are configured, you should see a console message on each router indicating that a full OSPF adjacency has formed, similar to the following sample output from IOU1:

```
%OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Ethernet0/0 from LOADING to FULL, Loading
Done
```

Task 3: Verify OSPF Neighbor Adjacency and Connectivity

On IOU1, issue the following command to confirm the OSPF neighbor adjacency:

```
IOU1#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
192.168.1.2 1 FULL/BDR 00:00:38 192.168.1.2 Ethernet0/0
```

From IOU1, ping IOU2's e0/0 interface (192.168.1.2):

```
IOU1#ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

On IOU1, issue the following command to confirm that the routing table reflects the OSPF-learned path (note: because 192.168.1.0/24 is directly connected on both routers in this simple topology, it will appear as a connected route rather than an OSPF route; in a larger topology, remote networks advertised by the neighbor would appear with the O flag):

```
IOU1#show ip route

Codes: C - connected, O - OSPF, S - static

C 192.168.1.0/24 is directly connected, Ethernet0/0
```

Because both pings succeed and the neighbor state shows FULL, the OSPF adjacency between IOU1 and IOU2 is confirmed to be working correctly.